INCH-POUND

MIL-I-81454(AS)
AMENDMENT 3
16 April 2001
SUPERSEDING
AMENDMENT 2
15 June 1970

MILITARY SPECIFICATION

INDICATOR, GYROSCOPE, VERTICAL REFERENCE
ID-1481 (*)/A

INACTIVE FOR NEW DESIGN AFTER 09 OCTOBER 1996

This amendment forms a part of MIL-I-81454(AS), dated 15 January 1967, and is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1, Under Specifications, Military: Add:

"MIL-PRF-6085	-	Lubrication Oil, Aircraft Instrument, Loss Volatility.
MIL-M-7793	-	Meter, Time Totalizing. (Inactive for New Design)
MIL-C-14806	-	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges. (Inactive for New Design)
MIL-C-26482	-	Connectors, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting. (Inactive for New Design)
MIL-I-81219	-	Indicator, Elapsed Time, Electro-Chemical. (Inactive for New Design)
MIL-PRF-81322	-	Grease, Aircraft, General Purpose, Wide Temperature Range
MIL-G-81559	-	Gyroscope Assemblies, Attitude and Directional Reference Instruments for Aircraft Preservation and Packaging, Procedures for. (Inactive for New Design)
MIL-B-81744	-	Barrier Coating Solution and Solvent"

2.1, Under Standard, Military: Delete "MIL-STD-794" and Title, Add "MIL-STD-1334 Barrier Coating 'Process for Anti-Friction Bearings" and "MIL-STD-838 Lubrication of Military Equipment"and"MIL-STD-471 Maintainability Demonstration."

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PAGE 3

* 3.1: Delete and substitute:

"3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3."

PAGE 4

3.3.2.3: Delete “3500” and substitute “2000.”

Add as new paragraphs:

"3.3.2.4 Time totalizing meter. The indicator shall contain a time totalizing meter (4,000 hours) conforming to MIL-M-7793. The reading of the meter shall not require unsealing of the indicator case. The time totalizer shall be installed within the present envelope of the instrument.

3.3.2.5 Indicator-elapsed time, electro-chemical. This indicator may be used as an alternate and shall conform to MIL-I-81219 within present envelope of the instrument. The scale shall be for 5,000 hours."

PAGE 5

3.3.9.1: Delete and substitute:

"3.3.9.1 Vibration. The indicator shall operate satisfactorily when subjected to the vibration tests of 4.8.18 and 4.8.19."

PAGE 8

Add as new paragraphs:

"3.5.1.2.5 Drum mask. The drum mask is affixed to rotate with the roll gimbal. Referring to figure 2, the portion of the drum mask above the drum shall be gray color No. 36440 of FED-STD-595."

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PAGE 10

3.5.1.7, Last two sentences: Delete and substitute:

"The warning flag shall be fluorescent red, color No.38905 of FED-STD-595 and the lettering "off" shall be black color No. 37038 of FED-STD-595. The fluorescent red flag shall be illuminated under night lighting conditions and the lighting shall not be entirely dependent upon one light source."

PAGE 11

3.5.1.10: Add as last sentence:

"This device shall not restrict the pitch and roll freedom in the absence of power."

3.5.1.11.3, Last sentence: Delete and substitute:

"The erection rate in pitch and roll shall be not greater than 0.9 degree per minute when the spin axis displacement is greater than 10 degrees in pitch and roll."

3.5.1.12: Delete and substitute:

"3.5.1.12 Rotor coast time. The indicator shall display pitch and roll information within 6 degrees of vertical for a minimum of 9 minutes after power is removed."

PAGE 12

Add as new paragraphs:

"3.6.1 Lubrication.

3.6.1.1 Rotor bearing lubrication. The gyro spin bearings shall be lubricated with grease MIL-PRF-81322 or MINIPURE grease or equivalent with particle size not greater than 50 microns. The requirements of MIL-STD-838 shall be met.

3.6.1.2 Bearing lubrication. The lubricating oil used in gimbal bearings shall meet the performance requirements of MIL-L-6085 except that it shall conform with Naval Research Laboratory polyolester lubricant NRL MB-20-B. Teflon separators shall be used in gimbal bearings.

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3.6.1.3 Barrier film bearing coating. All bearings lubricated in accordance with requirements of 3.6.1.1 and 3.6.1.2 shall be processed in accordance with MIL-STD-1334. The coating material shall be in accordance with MIL-B-81744. All bearings shall have chamfered races and shall be barrier film processed either by a ball bearing manufacturer or a designated Government facility.

3.7 Operations. Precessions in pitch and roll with rated power shall be in accordance with 3.7.1, 3.7.2, and 3.7.3.

3.7.1 Longitudinal acceleration. When subjected to an 8G acceleration for 4 seconds along the roll axis, the indicator shall not precess in pitch and roll beyond 2 degrees (see 4.8.9.2.3).

3.7.2 Coordinated turns. When subjected to coordinated turns with a turn rate between 90 to 1500 degrees per minute and bank angles of 20 to 60 degrees, the indicator shall not precess in pitch and roll beyond 2 degrees maximum (see 4.8.9.2.1 and 4.8.9.2.2).

3.7.3 Uncoordinated turns. When subjected to uncoordinated turns (skid turns) of less than 5 degrees per minute at air speeds not greater than 400 knots for a 10 minute duration, the indicator shall not process in pitch and roll beyond 2 degrees maximum."

4.1.1 and 4.2: Delete and substitute:

* "4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Conformance inspection (see 4.4)

4.2.1, 4.2.2, and 4.2.3: Delete and substitute:

* “4.3 First article inspection. The first article inspection shall consist of the Life test in 4.5 and all the tests in 4.8.

* 4.3.1 First article samples. First article inspection samples shall consist of 3 samples. The samples shall have been previously subjected only to individual inspections.

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* 4.3.1.1 First article sample identification The first article inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Samples for First Article Inspection
(Name of Equipment)
Submitted by (Manufacturer's name, date and
part number) for First Article Inspection in
accordance with MIL-I-81454(AS)
under authorization (reference letter authorizing tests)."

PAGE 14

4.3, 4.3.1, 4.3.2, 4.3.3, and 4.4: Delete and substitute:

"4.4 Conformance inspection. The contractor shall furnish all samples and shall be responsible for performing all tests except sampling plan B tests which will be conducted at a government laboratory designated by the procuring activity. Conformance inspection shall consist of the following:

(a) Individual tests
(b) Sampling plan A tests
(c) Sampling plan B tests
(d) Reliability assurance tests
(e) Special tests"

PAGE 15

4.4.2: Delete and substitute:

"4.4.2 Sampling plans. The sampling plans shall consist of sampling plan A and sampling plan B tests. The test samples selected shall first have passed the Individual test.

Quantity Offered for Acceptance	Quantity to be Selected for Sampling Plan A Tests
First 15	1 (zero when Sampling Plan B is invoked)
Next 50	1
Each 100 of remainder	1"

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4.4.2.1: In the title and first sentence change "sampling tests" to read "sampling plan A tests."

Add as new paragraphs:

"4.4.2.2 Sampling plan B instructions. One sample shall be selected at random from the first 15 produced on contract and submitted within ten days after manufacture. These samples shall be forwarded at the contractor's expense to a Government laboratory designated by the procuring activity. Each sample shall be identified by a securely attached durable tag marked with the following information:

Indicator Gyro ID-____/A
Submitted by (Manufacturer's name, date) for
Production Acceptance Sampling Plan B Inspection
of MIL-I-81454 with Contract No. _____
Manufacturer's Part No. _____

4.4.2.2.1 Sampling plan B approval. Approval of sampling plan B instruments shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall cause cessation of deliveries until corrective action is performed.

4.4.2.2.2 Sampling plan B inspection. Each sample selected for sampling plan B testing shall be subjected to the following:

(a) Sampling Plan A Tests
(b) Vibration Failure
(c) Radio Noise Interference
(d) High Temperature Exposure
(e) High Altitude - Low Temperature
(f) Temperature Shock
(g) Humidity
(h) Salt Spray
(i) Fungus Resistance
(j) Impact Shock
(k) Acceleration
(1) Sand and Dust"

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PAGE 17

4.4.3.1: Delete.

4.4.3.2 and 4.4.3.2.1: Delete and substitute:

* "4.4.3.2 Reliability production acceptance phase. Reliability production acceptance testing shall be conducted through a production burn-in. Each equipment produced but not submitted for the Reliability First Article Test shall be operated for 150 operating hours burn-in, and the test data (operating hours and failures) shall be compared to the reject criteria of test plan V of MIL-STD-781. (Extend the lines as necessary to accommodate the data omitting the truncation lines). Delivery of equipment shall not be held up to await a decision under the test plan. At any time that the current total of test hours and test failures plotted on test plan V curves show a reject situation, the procuring activity shall be notified. In the event the total of test hours and test failures plotted on test plan V lines extended shows an accept situation, the test duration on each subsequently produced indicator shall be reduced to 100 operating hours. For the production acceptance phase, test level F shall be used, except that the temperature extremes shall be +55°C and -44°C, and vibration shall be 0.5G."

PAGE 19

4.5, Third sentence: Delete "Preproduction" and substitute "First Article."

PAGE 21

4.6, First sentence: Delete "Preproduction" and substitute "First Article."

PAGE 23

4.8.7: Add as last sentence: "The test shall be repeated with power removed."

PAGE 24

4.8.9.2.1, Second sentence: Delete and substitute:

"The roll and pitch axis shall not change greater than 3.5 degrees in a 4 minute period."

4.8.9.2.2, Last sentence: Delete and substitute:

"The roll and pitch indication shall not change greater than 3.5 degrees in a 4 minute period."

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Add as new paragraph:

"4.8.9.2.3 Vertical angle erection. With the roll axis of the indicator oriented along the vertical and the spin axis oriented towards the geographical north, the gyro shall be aligned to indicate 0 degrees in pitch and roll. The roll and pitch indication shall not change greater than 2 degrees in a 32 second period."

PAGE 25

4.8.12: Delete and substitute:

"4.8.12 Rotor coast time. The pitch and roll indication shall not change greater than 6 degrees in a 9 minute period."

4.8.14, First sentence: Delete and substitute:

"The indicator shall be connected and subjected to the transient variations specified in MIL-STD-704. No damage shall occur to the indicator. The indicator shall then be tested with the input voltage and frequency being varied within the following limits and combinations."

PAGE 26

4.8.18, After second sentence: Add: "Linear motion may be used instead of circular motion."

PAGE 27

4.8.19: Delete and substitute:

"4.8.19 Vibration failure. The indicator shall be tested in accordance with Part II, Vibration Test of MIL-T-5422 except the following vibration curve shall be substituted in lieu of Curve II of figure 2: between 5 Hz and 20 Hz a constant double amplitude of 0.10 inch shall be applied, between 20Hz and 350 Hz a constant 2G vibration shall be applied, between 350 Hz and 2,000 Hz a constant 5G vibration shall be applied. Power, including power to the lighting circuit, shall be applied to the unit during the vibration test. Following exposure to vibration the unit shall meet the individual test requirements."

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PAGE 28

Add as new paragraphs:

"4.8.28.1 Operations. Rated power shall be applied to the equipment and precessions in pitch and roll resulting from accelerations shall be limited as stated in 3.7.

4.8.28.1.1 Longitudinal accelerations. The equipment shall meet the requirements of 3.7.1.

4.8.28.1.2 Coordinated turns. The equipment shall meet the requirements of 3.7.2.

4.8.28.1.3 Uncoordinated turns. The equipment shall meet the requirements of 3.7.3."

PAGE 29

4.8.30, Third sentence: Delete and substitute:

"The error in either pitch or roll shall be not greater than three degrees. The error induced by roll maneuvers at a rate of 300° ± 10° per second shall be not greater than one degree. Furthermore, rotating the indicator in either direction at the above rates shall not cause damage to the indicator."

Add as new paragraph:

"4.8.31 Sand and dust. The indicator shall be subjected to Sand and Dust Tests, Procedure I of MIL-E-5272. At the end of sand exposure, the indicator shall not be damaged and shall meet the requirements of the Individual tests."

PAGES 29 and 30

5.1, 5.2, 5.3, 5.4, and 5.4.1: Delete and substitute:

"5.1 Packaging, Packing, Preservation and Marking. Shall be in accordance with MIL-G-1559."

PAGE 32

6.2: Delete "6.8 Ordering data" and substitute "6.2 Ordering data."

Add as 6.2 (f):

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* "(f) When first article is required (see 3.1)."

* 6.6: Delete.

Add as new paragraph:

"6.7 Maintainability demonstration. Maintainability demonstration shall be in accordance MIL-STD-471 except as modified herein. If the demonstration has been performed on a previous contract or order, it need not be repeated."

Add as new paragraph, but delete "6.8" and substitute "6.2" (see 6.2 above).

* "6.2 Ordering Data. Procurement documents should specify the following:

(a) Title, number, and date of specification.

(b) The quantity, US part number and type number, if any, of the instrument desired.

(c) Levels of packaging and packing desired.

(d) The laboratory that shall conduct tests.

(e) Reliability test phases (See 4.7).

(f) When first article is required (see 3.1)."

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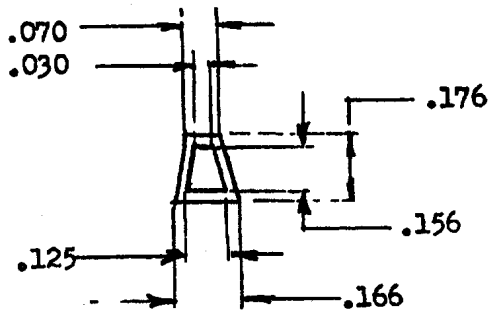
Figure 5: Delete and substitute the new Figure 5 below.



All dimensions in inches
Tolerances for decimals ± 0.005

FIGURE 5. Bank index.

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